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To the Adjudicating Officer,

Please accept this letter as an expression of my wholehearted support for Dr. Akul Yugesh Mehta and his continued biotechnologies research in the United States. Dr. Mehta's leadership in biotechnologies is clearly demonstrated by his creation of multiple technologies that are invaluable for studying glycan protein interactions. In addition, Dr. Mehta's expertise in glycan microarrays and glyco-bioinformatics has made him an important member of the MIRAGE (Minimum Information Required For A Glycomics Experiment) Commission, and he has been invited as a panelist for various international conferences worldwide. There is thus no doubt that Dr. Mehta shall continue to produce additional valuable research, both driving progress in the field and bringing immense benefits to national and international concerns.

I wish now to establish my relevant qualifications as a recommender for Dr. Mehta. Firstly, I hold a Ph.D. in glycoproteomics from the University of Denmark. I specialize in analytical and structural glycobiology with a focus on the involvement of particular proteins in cellular homeostasis and diseases. Presently, I am an associate professor at Macquarie University and a visiting professor at Nagoya University in Japan. I have worked for Macquarie University since 2010 in a number of positions, including as the Cancer Institute NSW Early Cancer Research Fellow and as a senior research fellow. I am also the chair of the Human Glycoproteomics Initiative (HUPO). Despite never meeting Dr. Mehta, I am very familiar with his research as I have used it as a critical citation in my published work. Therefore, I am able to provide an informed recommendation of his work without any inherent bias or agenda.

In a study that has informed my own research, Dr. Mehta focused on glycans, a type of carbohydrate polymer that is important for various biological systems such as host-pathogen recognition, immune reactions, and transfusion medicine. Purified glycans are rare and extremely expensive, yet hundreds of glycans are required to study their biological interactions. Dr. Mehta explored different methods for developing an efficient use of these compounds, one of which included microarrays where picograms of several glycans and glycoconjugates are attached to a specially coated glass slide. These slides are used to test biological samples, where they provide binding data for a number of glycans in a rapid, high-throughput way. Notably, Dr. Mehta applied this method to generate vast glycan microarray libraries, which represents an entirely novel use that highlights the pioneering nature of his research efforts. Indeed, Dr. Mehta produced over 7000 arrays of over 20 types with over 500 glycans and glycopeptides, offering the scientific community a great resource for generating an enormous amount of data that has already been used in academic research and quality control procedures by companies in the United States and around the world. For example, these microarrays have been made available for use by hundreds of scientists

nation- and worldwide through the National Center for Functional Glycomics at Beth Israel Deaconess Medical Center, where Dr. Mehta is employed. Throughout this project, he provided his fellow researchers with a foundational understanding of many previously unknown glycan interactions.

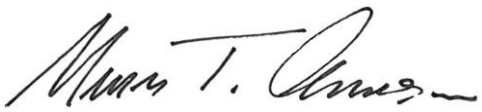
Dr. Mehta's second project led my co-authors and I to cite his findings in our paper titled "Trends in oligomannosylation and α1,2-mannosidase expression in human cancers." In our paper, we directly state that Dr. Mehta's newly developed microarrays open up new opportunities for the field to identify additional receptors that are present on cancer cells and tissues. We noted that the technologies created by Dr. Mehta are vital for the future of cancer diagnostics and therapeutic research in the United States and around the world.

Another of Dr. Mehta's significant studies is his work building GLAD, a web application tool for conducting glycan microarray data analysis and visualization. He created GLAD using CSS, Javascript, and HTML with the data visualization library D3js. GLAD allows researchers to visualize data in their web browser from numerous glycan array experiments and compare them using different data visualizations including heatmaps, force graphs, and bar charts. Furthermore, GLAD boasts data-wrangling capabilities that help researchers delve deeper into data and uncover patterns, and it is the only platform for glycan microarray data analysis that is used in a browser without having to upload data to a web server. This allows users to easily access this tool and use it for their own glycan microarray data analysis—indeed, since its inception, Dr. Mehta's GLAD tool has been used more than 4,000 times, and over the span of just one year, the tool was used over 1,500 times by other researchers in the field of biotechnologies.

These two research projects highlight Dr. Mehta's tremendous skills as a biotechnology researcher, as they show the importance of his novel glycan microarrays and techniques. This technology allows global research teams to research and understand novel pathways for HIV, COVID-19, and rheumatoid arthritis, which is critical for developing new therapies. Each of these conditions is debilitating, and Dr. Mehta's work helps teams study these diseases and identify new drug targets. As for GLAD, this tool allows scientists to easily access data from glycan experiments and compare them against existing data. By making his software publically available on the internet, Dr. Mehta is allowing every scientist to advance their glycan microarray studies.

To close, Dr. Mehta's impressive publication record in prominent journals serves as a testament to his outstanding work in biotechnologies. His research has been published in reputable journals, such as *Frontiers in Immunology, Molecular & Cellular Proteomics, Cell Chemical Biology,* and *Scientific Reports*. These journals only accept the highest quality of work, so their selection of Dr. Mehta's work evidently denotes his position as a prolific researcher in biotechnologies.

Respectfully yours,



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